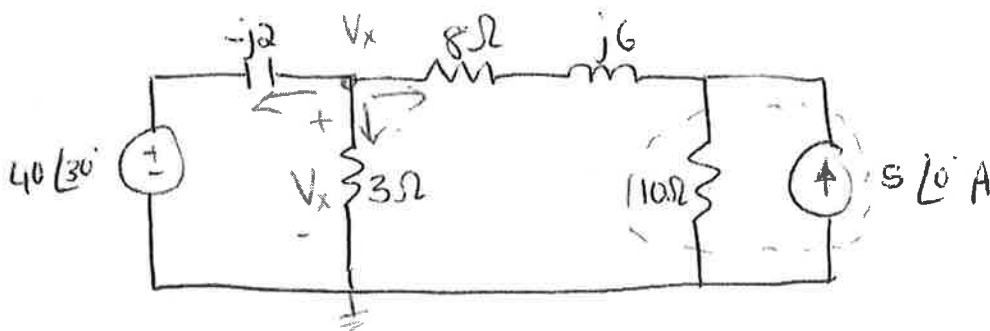


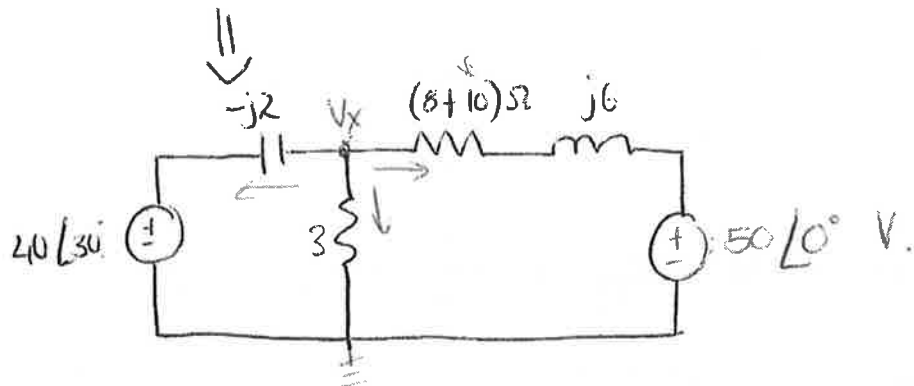
Tutorial 9

①

10.13.



Determine V_x .



* Nodal analysis:

$$\frac{V_x - 40\angle 30^\circ}{-j2} + \frac{V_x}{3} + \frac{V_x - 50\angle 0^\circ}{18 + j6} = 0$$

$$j0,5(V_x - 40\angle 30^\circ) + 0,33V_x + \frac{V_x}{18,97\angle 18,4^\circ} - 2,635\angle -18,4^\circ = 0$$

$$j0,5V_x + 0,33V_x + (0,05 - j0,0166)V_x = 2,5 - j0,8317 - 10 + j17,32j$$

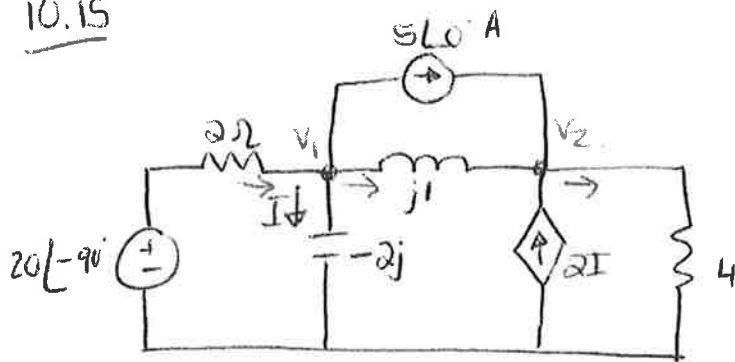
$$(0,38 + j0,4834)V_x = -7,5 + j16,4883$$

$$V_x = 29,4\angle 62,8^\circ$$



10.15

(2)



Solve for I using nodal analysis.

@ Node 1 :

$$\frac{20\angle-90^\circ - V_1}{2} = 5\angle 0^\circ + \frac{V_1}{-2j} + \frac{V_1 - V_2}{j1}$$

$$\frac{-20j - V_1}{2} = 5 + \frac{V_1}{-2j} + \frac{V_1 - V_2}{j1}$$

$$-10j - 0,5V_1 = 5 + j0,5V_1 - jV_1 + jV_2$$

$$-5 - 10j = (0,5 - 0,5j)V_1 + jV_2 \quad \text{--- (1)}$$

@ Node 2 :

$$\frac{V_1 - V_2}{j1} + 5\angle 0^\circ + 2I = \frac{V_2}{4}$$

$$* \boxed{I = \frac{V_1}{-2j}}$$

$$-jV_1 + jV_2 + 5 + jV_1 = 0,25V_2$$

$$V_2 = \frac{5}{0,25 - j} \quad \text{--- (2)}$$

$$\Rightarrow \text{in (1)}: -5 - j10 = 0,5(1 - j)V_1 + j\left(\frac{5}{0,25 - j}\right)$$

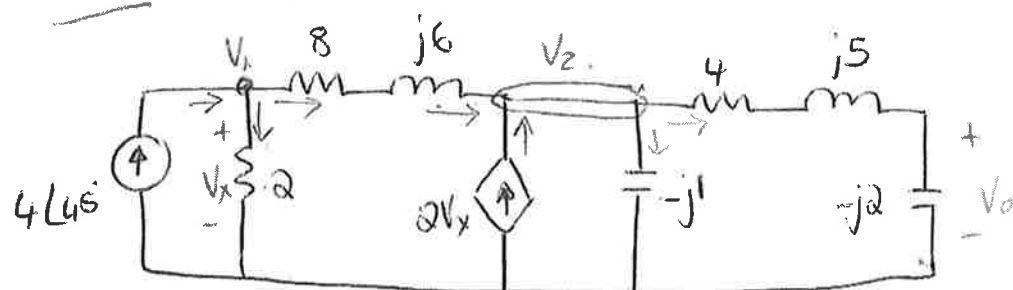
$$(1 - j)V_1 = -10 - j20 - \left(\frac{j40}{1 - j4}\right)$$

$$V_1 = 15,81 \angle -46,5^\circ$$

$$I = \frac{V_1}{-2j} = 7,906 \angle 43,49^\circ$$

(3)

10.18 *

Use nodal analysis to obtain V_o .

* ①

$$4\angle 45^\circ = \frac{V_1}{2} + \frac{V_1 - V_2}{8 + j6}$$

$$200\angle 45^\circ = (29 - j3)V_1 - (4 - j3)V_2 \quad \text{--- ①}$$

* ②

$$\frac{V_1 - V_2}{8 + j6} + 2V_x = \frac{V_2}{-j1} + \frac{V_2}{4 + j5 - j2}$$

$$V_1 = \left(\frac{12 + j41}{104 - j3} \right) V_2 \quad \text{--- ②}$$

$$* \boxed{V_x = V_1}$$

Substitute ② in ①

$$\Rightarrow 200\angle 45^\circ = (29 - j3) \left(\frac{12 + j41}{104 - j3} \right) V_2 - (4 - j3)V_2$$

$$\therefore V_2 = \frac{200\angle 45^\circ}{14.21\angle 89.17^\circ} = 14.07\angle -44.17^\circ$$

$$\therefore V_o = \left[\frac{-j2}{4 + j5 - j2} \right] V_2$$

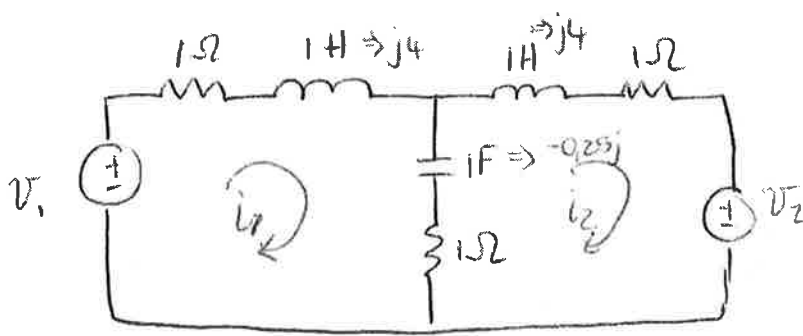
$$= 0.4\angle -126.87^\circ V_2$$

$$= 5.63\angle -171.03^\circ \Rightarrow 5.63\angle 360^\circ - 171.03^\circ$$

$$= 5.63\angle 188.97^\circ$$

10.28

4



Determine i_1 and i_2 .

Let $V_1 = 10 \cos 4t \text{ V}$

$V_2 = 20 \cos(4t - 30^\circ) \text{ V}$

* $1 \text{ H} \Rightarrow j\omega L = j4$

$1 \text{ F} \Rightarrow \frac{1}{j\omega C} = -j0.25$

* $V_1 = 10 \angle 0^\circ$

* $V_2 = 20 \angle -30^\circ$

Mesh 1:

$$-10 + (2 + 3.75j)I_1 - (1 - 0.25j)I_2 = 0$$

$$(2 + 3.75j)I_1 = 10 + (1 - 0.25j)I_2 \quad \text{--- (1)}$$

Mesh 2: $-(1 - 0.25j)I_1 + (2 + 3.75j)I_2 = -20 \angle -30^\circ \quad \text{--- (2)}$

$$\Rightarrow \begin{bmatrix} (2 + 3.75j) & -(1 - 0.25j) \\ (-1 + 0.25j) & (2 + 3.75j) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 10 \\ -17.32 + j10 \end{bmatrix}$$

$$\Rightarrow I_1 = 2.741 \angle -41.07^\circ$$

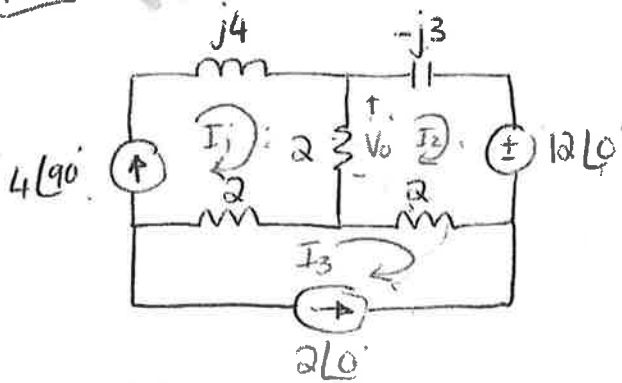
$$I_2 = 4.114 \angle 92^\circ$$

$$i_1(t) = 2.741 \cos(4t - 41.07^\circ) \text{ A}$$

$$i_2(t) = 4.114 \cos(4t + 92^\circ) \text{ A}$$

10.36

Compute V_o using mesh analysis.



$$\rightarrow I_1 = 4 \angle 90^\circ = 4j$$

$$\rightarrow I_3 = -2$$

* From mesh 2:

$$-2I_1 + (4 - j3)I_2 - 2I_3 + 12 = 0$$

$$(4 - j3)I_2 = 8j - 2 - 12$$

~~$$I_2 = \frac{-24 + j8}{4 - j3}$$~~

$$I_2 = \frac{-16 + j8}{4 - j3}$$

$$= -3,52 - j0,64$$

$$\therefore V_o = 2(I_1 - I_2)$$

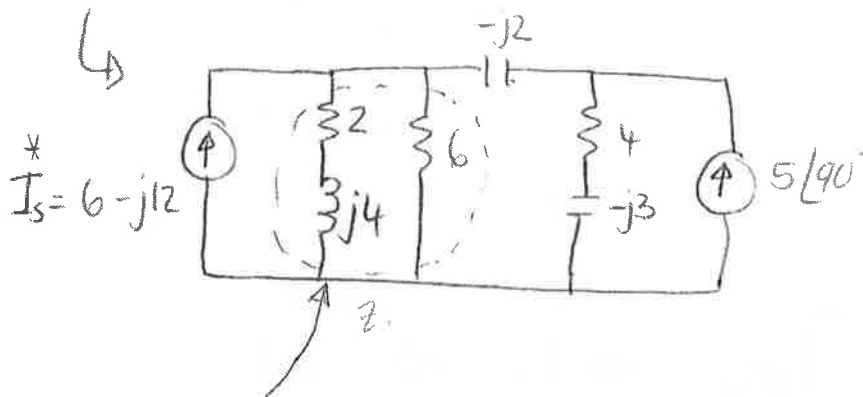
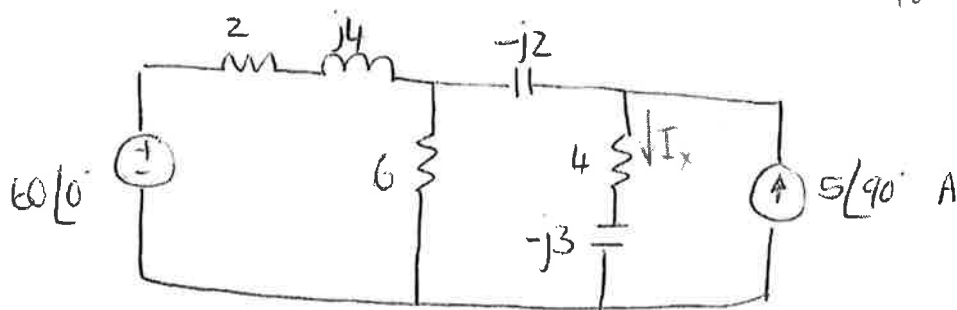
$$= 2(j4 + 3,52 + j0,64)$$

$$= 7,04 + j9,28$$

$$= 11,648 \angle 52,82^\circ \text{ V}$$

10.52

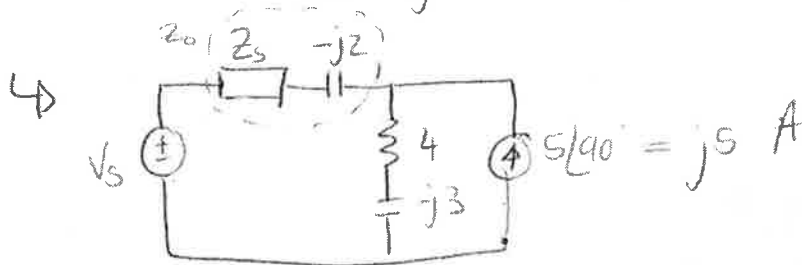
Use source transformation to find I_x



$$* I_s = \frac{60 \angle 0^\circ}{2 + j4} = (6 - j12) \text{ A}$$

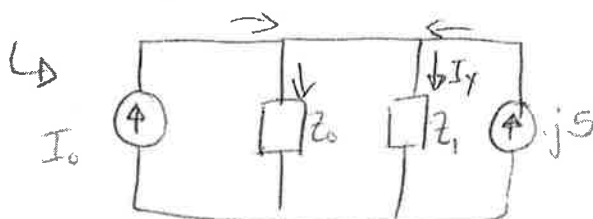
$$\therefore Z = 6 \parallel (2 + j4) = 2,4 + j1,8$$

$$V_s = I_s Z = (6 - j12)(2,4 + j1,8) = 36 - j18$$



$$\therefore Z_0 = Z_s - j2 = 2,4 - j0,2$$

$$\therefore I_0 = \frac{V_s}{Z_0} = 15,517 - j6,207$$



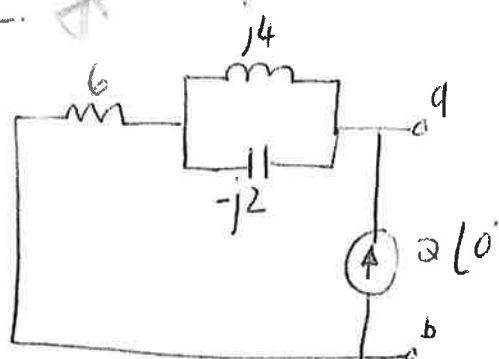
$$I_x = \frac{Z_0}{Z_0 + Z_1} (I_0 + j5) = 5 + j1,563 = (5,238 \angle 17,35^\circ) \text{ A}$$

10.56

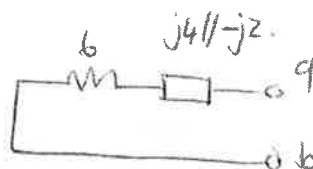
Thevenin & Norton.

(7)

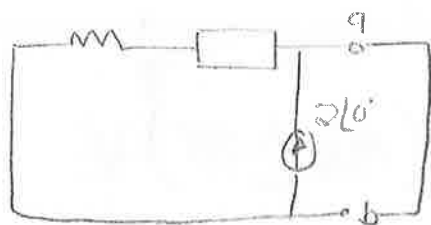
a)



$$\textcircled{*} Z_N = Z_{TH} = 6 + (j4 \parallel -j2) \\ = (7,22 \angle -33,69^\circ) \Omega.$$



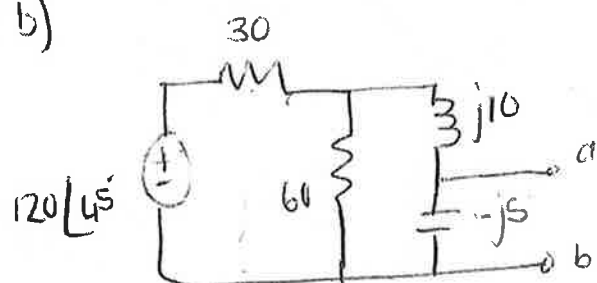
\textcircled{*}



$$\Rightarrow I_N = 20 \angle 0^\circ \text{ A.}$$

$$\therefore V_{TH} = I_N (Z_{TH}) = (14,422 \angle -33,69^\circ) \text{ V}$$

b)



$$30 \parallel 60 = 20.$$

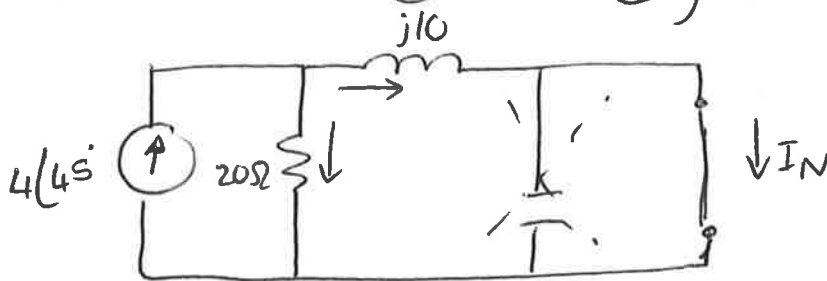
$$\textcircled{*} Z_{TH} = Z_N \\ = -j5 \parallel (20 + j10) \\ = 5,423 \angle -77,47^\circ$$



stop.

⊗ V_{TH} or I_N

(transform $\oplus \rightarrow \oplus$)



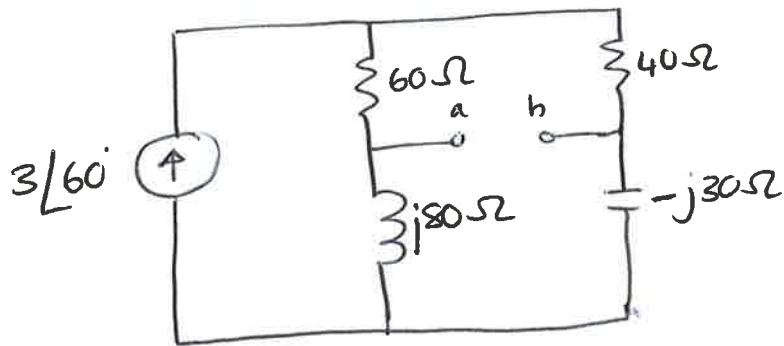
$$\therefore I_N = \left(\frac{20}{20 + j10} \right) 4\angle 45^\circ$$

$$= (3,578 \angle 18,43^\circ) \text{ A}$$

$$\therefore V_{TH} = I_N Z_N = (19,4 \angle -59^\circ) \text{ V.}$$

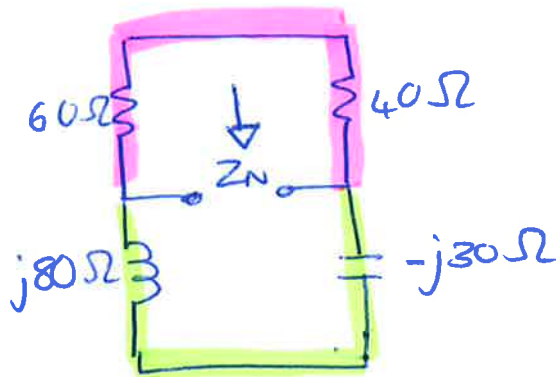
10.64

(9)



Find the NORTON-equivalent at terminals a-b.

Z_N

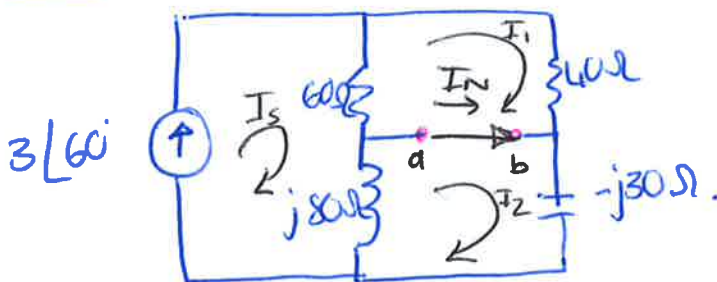


$$\begin{aligned} Z_N &= (60 + 40) \parallel (j80 - j30) \\ &= 100 \parallel j50 \\ &= 20 + j40 \\ &= 44.72 \angle 63.43^\circ \end{aligned}$$

$$\begin{aligned} &* 100 \parallel j50 \\ &= \frac{(100)(j50)}{100 + j50} \end{aligned}$$

→

I_N



• $I_s = 3 \angle 60^\circ$

• Mesh 1:

$$100I_1 - 60I_s = 0$$

$$I_1 = 1.8 \angle 60^\circ$$

• Mesh 2:

$$(j80 - j30)I_2 - j80I_s = 0$$

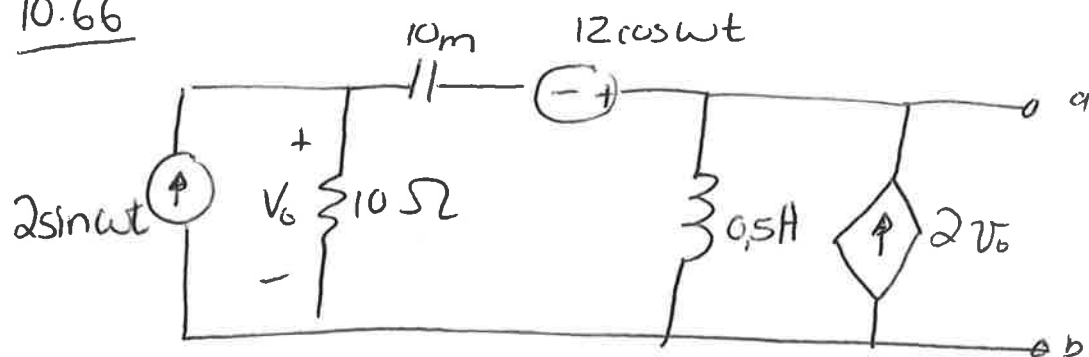
$$I_2 = 4.8 \angle 60^\circ$$

$$I_N = I_2 - I_1$$

$$= 3 \angle 60^\circ$$

→

10.66

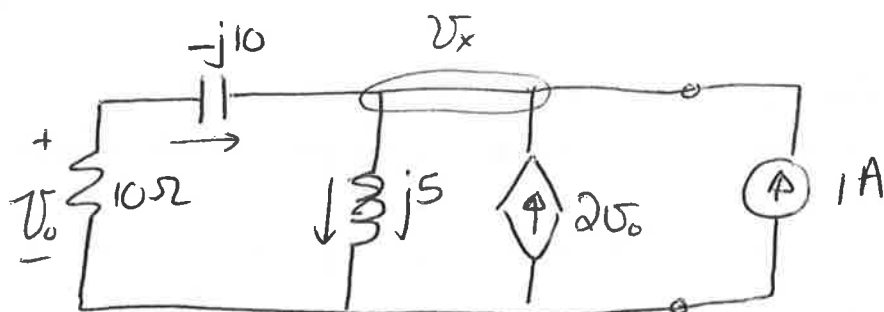


* Thevenin
Σ Norton

10

* $Z_{TH} = Z_N$ (Dependant source $\Rightarrow$ TEST SOURCE)

($\omega = 10 \text{ rad/s}$)



* $V_o = V_x \left(\frac{10}{10-j10} \right) \rightarrow \text{voltage division.}$

* @ node x:

$$\frac{V_x}{10-j10} + \frac{V_x}{j5} = 2V_o + 1$$

$$\frac{V_x}{10-j10} + \frac{V_x}{j5} - 2\left(\frac{10}{10-j10}\right)V_x = 1$$

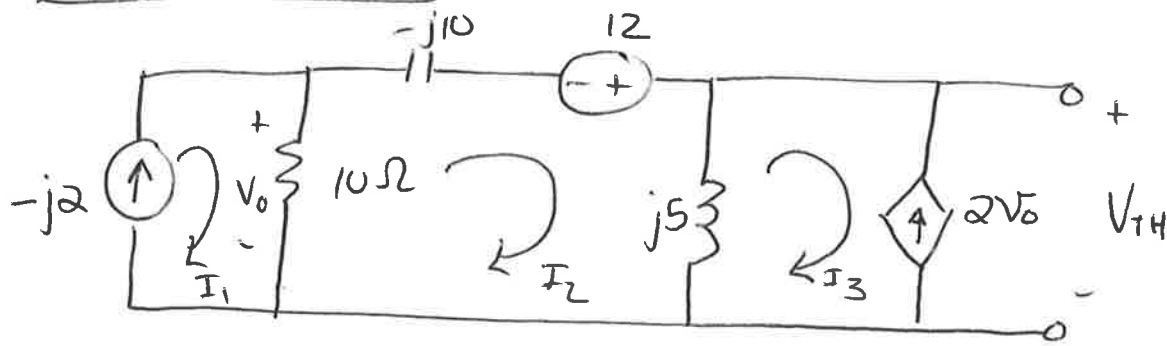
$$\therefore V_x = \frac{-10+j10}{21+j2}$$

$$\therefore Z_N = Z_{TH} = \frac{V_x}{1}$$

$$= 0,670 \angle 129,56^\circ \Omega$$

V_{TH} and I_N

(15)



* $I_1 = -j2$

* $I_3 = 2V_0$

* Loop 2:

$$-10(-j2) + (-j10)I_2 - 12 + j5I_2 - j5I_3 = 0$$

* $V_0 = 10(I_1 - I_2)$

* $j5(2V_0)$

$$-10(-j2) + 10I_2 - j10I_2 + j5I_2 - j10[-20j - 10I_2] = 12$$

$$(10 - j105)I_2 = 12 - j20 - 200$$

$$I_2 = 0,0197 - j1,7924$$

$$\begin{aligned} \therefore V_{TH} &= j5(I_2 - I_3) \\ &= (29,78 \angle -3,615^\circ) V \end{aligned}$$

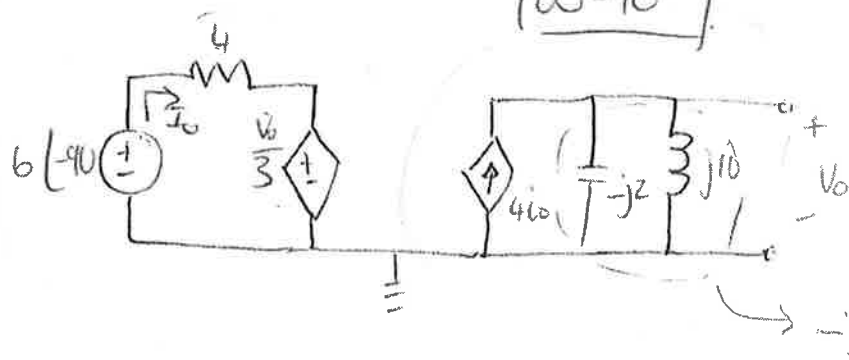
$$\begin{aligned} V_0 &= 10(-j2 - 0,0197 + j1,792) \\ &= -0,197 - j2,0764 \end{aligned}$$

$$\begin{aligned} \therefore I_N &= \frac{V_{TH}}{Z_{TH}} \\ &= (44,46 \angle -133,16^\circ) A \end{aligned}$$

10.68 *

$\omega = 10$

Thevenin



V_{TH} :

• $V_0 = V_{TH} = 4I_0(-j2,5) = -j10I_0$ — (1)

• $-6\angle-90 + 4I_0 + \frac{V_0}{3} = 0$ — (2)

① in ② • $j6 + 4I_0 + \frac{-j10I_0}{3} = 0$

$(4 - j3,33)I_0 = -j6$

$I_0 = 0,738 - j0,885$

• $V_{TH} = -j10I_0$
 $= 11,52 \angle -140,197^\circ$

• $V_{TH} = 11,52 \sin(10t - 50,197^\circ)$

Z_{TH} :

• $4I_0 + \frac{V_0}{3} = 0$
 $I_0 = -\frac{V_0}{12}$ — (1)

• $1 + 4I_0 = \frac{V_0}{-j2} + \frac{V_0}{j10}$ — (2)

① in ② $Z_{TH} = 1,2293 - j1,4766$

• $Z_{TH} = \frac{V_0}{I} = (1,2293 - j1,4766) \Omega$

